

PTC Uniform/Constant Analysis Coefficient

Scientific Rationale and Contract

SCI-PTC-COEFFICIENT-UNIFORM v0.1-draft.1 / r0.1

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Abstract

Draft for scientific-owner review. This view explains a complete proposal for a dimensionless uniform PTC analysis/gridding coefficient. Every new numerical, profile, permission, and boundary choice remains unapproved. The document claims algebraic definition only; it claims no implementation conformity, validation, performance, readiness, registered family, policy, or default.

1 Program adherence and prior-work recovery

This implementation-blind first scientific draft follows the released Scope Brief, its completed prior-work recovery synthesis, and the admitted program charter excerpts. It adopts the frozen PTC coefficient taxonomy, generic coefficient interface, immutable lifecycle, named-use separation, and PTC-disabled route; the approved single-registry framework; the frozen MAP and JINC coefficient-boundary constraints; the frozen NOI balance and lossless rational-parent obligations; and the admitted VAL knowledge, binding, and four-axis rules. It cites the generic PTC construction instead of re-deriving it. Implementation, tests, audits, repairs, validation, and external scientific material were excluded.

The genuinely new content is one proposed uniform/constant family and one complete proposed PTC-owned coefficient-handoff profile. The family, profile, registry key, two boundary successors, numerical definition, normalization, support, permissions, and policies are proposals awaiting scientific-owner review. No request default or mode policy is proposed. Sensitivity, residual-scatter, hybrid, learned, and adaptive families remain deferred.

Compact input/output/equation/source/status block

Input	One exact realized PTC transformed product Z ; its immutable CAL parent; application generation; observation and occurrence identities; output support; requested/effective/resolved/realized coefficient selection; and preserved typed facts.
Output	A proposed PTC-owned positive analysis/gridding coefficient ω_i (also denoted γ_i at the MAP boundary), its immutable family generation and payload, and a complete named-use decision record per requested occurrence and named consumer.
Equation	$\tilde{\omega}_i = 1$ and $\omega_i = \tilde{\omega}_i / (\mathcal{D} ^{-1} \sum_{j \in \mathcal{D}} \tilde{\omega}_j) = 1$ for every $i \in \mathcal{D}$, with $\mathcal{D}$ exact and nonempty.
Sources	Released charter/scope/recovery material; frozen PTC, MAP, JINC, and NOI excerpts; approved PTC registry framework; admitted VAL Core, Registry, and PTC common-fragment excerpts.
Status	Algebraically complete author proposal. Owner adoption, implementation conformity, representation fidelity, observational performance, and production readiness are not established.

Symbols

o is one observation; p is one exact realized PTC transformed-product and application generation; g is one immutable coefficient generation bound to p ; i is one exact detector-time occurrence; ι_i is its complete stable identity; $\mathcal{S}_{p,o}^{\text{PTC,out}}$ is the exact PTC output support; $\mathcal{D}_{g,o}$ is the coefficient normalization population; $\tilde{\omega}_i$ and ω_i are the raw and normalized

PTC coefficients; γ_i is MAP's notation for the same PTC value; and κ_{ip} is the separately JINC-owned signed spatial factor.

2 Physical meaning and intended use

The family assigns equal PTC coefficient influence to every occurrence in one explicitly declared coefficient support. It is a transparent baseline: relative differences in downstream contribution cannot originate from this PTC coefficient. Other producer and consumer factors can still make total contributions unequal. MAP retains its own admission, placement, boundary, exposure, and contribution rules. JINC retains its signed spatial factor κ_{ip} and its own normalization, conditioning, and support.

Uniformity is a statement only about the PTC-owned positive coefficient. It does not assert equal signal quality, equal exposure, equal response, equal uncertainty, detector independence, inverse variance, sensitivity, significance, precision, or optimality. It does not restore an invalid sample, create output support, or override any consumer decision.

3 Common registry-entry contract

Facet	Proposed exact content
Registry/family identity	SCI-PTC:analysis_gridding_coefficients@draft-0.1; family SCI-PTC:uniform_constant@draft-0.1; scientific owner SCI-PTC. The proposed actual human owner binding is Grant Wilson acting for SCI-PTC; exact confirmation is an open owner decision. Both identities remain drafts until owner adoption and immutable Registry binding.
Index and broadcast	Logical detector-time occurrence i with full exact identity. One scalar may represent the family only through the lossless broadcast in Equation 1 and its bound member identities.
Statistic and factors	No measured statistic and no fitted, residual, sensitivity, exposure, response, covariance, or loading factor. The sole raw factor is fixed $c_\star = 1$.
Unit	Dimensionless unity. The unit conveys no precision or uncertainty meaning.
Normalization	Named operator $\mathcal{N}_{\text{mean1}}$ in Equations 5–6; exact population $\mathcal{D}_{g,o}$ in Equation 2.
Estimation population	None: the value is defined rather than estimated. The normalization population still identifies its exact scientific scope.
Support	Exactly the bound product's PTC output support for one observation. Fit and application supports remain separate producer facts.
Parent compatibility	Exact PTC transformed product/application generation, immutable CAL parent, observation, occurrence identities, and selection lineage must match. Numerical equality establishes no compatibility.
Uncertainty meaning	The coefficient has no sampling uncertainty because it is an exact policy constant. This supplies no zero for signal, calibration, response, covariance, selection, representation, or performance uncertainty; those states remain separate.
Lifecycle	Requested, effective-policy, observation-resolved, selected, realized, published, and evaluation states are immutable and one-way linked. A re-realization creates a new g and never modifies p or an earlier g .
Consumer permissions	Separate proposals to permit SCI-MAP and SCI-JINC. Each requires owner adoption. Neither can be inferred from the other.
Numerical use	Positive multiplicative analysis/gridding coefficient in already-governed MAP or JINC consumer arithmetic.
Prohibited meanings	Precision, inverse variance, sensitivity, significance, fitted loading, residual scatter, validity, exposure, response, covariance, independence, optimality, default, support restoration, or missing-value substitution.
Missing/invalid/conflict	Missing selection, family, binding, support, payload, profile decision, or permission makes the affected route unavailable. Producer-invalid payload is not consumed. Structural conflict makes the decision unavailable. No state becomes zero or unity by fallback.

Facet	Proposed exact content
Unrepresentable state	If a representation cannot encode exact dimensionless one and its exact broadcast identity, the payload is typed unrepresentable/unavailable and the generation cannot be claimed realized.

4 Proposed complete coefficient-handoff profile

4.1 Identity, use, and object

The proposed profile is `SCI-PTC:uniform_coefficient_handoff@draft-0.1`. Its scientific package/producer owner is SCI-PTC, and its proposed actual-owner field is Grant Wilson acting for SCI-PTC, pending exact confirmation. Its exact use is:

For one requested tuple (g, o, i, c) , decide whether a realized PTC coefficient payload is a faithful instance of `SCI-PTC:uniform_constant@draft-0.1` and is authorized for handoff to the explicitly named consumer $c \in \{\text{SCI-MAP}, \text{SCI-JINC}\}$.

The object is one coefficient occurrence plus its generation-level scalar and broadcast binding. The profile binds the admitted PTC common named-use fragment and the exact eventual owner-approved source bytes of this contract. It is not the unsupported generic name `SCI-PTC:coefficient_qc_population@1`; it is a complete narrower proposition. While its source or Registry record remains draft or absent, binding integrity is not authoritative and no eligible evaluation may be claimed.

4.2 Applicability and structural gate

The profile is requested only when the exact family is the resolved selection for a named consumer and a coefficient handoff is requested. It is applicable when the exact PTC product exists, $\mathcal{D}_{g,o}$ is authoritatively nonempty, and $i \in \mathcal{D}_{g,o}$. It is inapplicable when authoritative identity establishes $i \notin \mathcal{D}_{g,o}$. It has `applicability_unknown` when product, observation, support, membership, or generation binding is missing or conflicting.

The nonexceptionable structural gate requires an active exact Registry record; actual owner and source binding; complete occurrence, observation, product, parent, family, generation, selection-lineage, support, scalar, broadcast, profile, and named-consumer identities; and consistent scopes and immutable provenance. Duplicate or ambiguous occurrence identity, inferred ordering, cross-observation pooling, or cross-generation pairing fails the gate and makes the decision unavailable rather than use-specifically ineligible.

4.3 Exact profile inputs and truth-domain rules

The profile consumes only the following owner-bound inputs after the structural gate establishes their identities: the resolved family-selection record and its parameter field; the PTC compatibility permission for the already-bound product and coefficient generations; the immutable PTC output-support record and exact member identities; the family-definition record naming constant, unit, operator, and population; the typed payload-availability record, published scalar bytes, declared decoder, unit, broadcast member binding, and publication digest; the exact named-consumer permission; and immutable lifecycle links and overwrite/nonmutation attestations. Sample validity, retention, consumer admission, response, covariance, and diagnostics are preserved inputs for provenance but are not permission predicates.

For each restriction below, **T** means its complete authoritative input set positively satisfies the stated predicate; **F** means the complete authoritative input set explicitly violates it; **U** means a required input or complete assertion is missing or the check was not performed; and **C** means authoritative inputs are contradictory, ambiguous, or outside the predicate's declared domain. A parent, occurrence, source, or generation identity mismatch is handled earlier by the structural gate and is therefore decision unavailable. It is not converted to **F** here.

The payload-fidelity input in UQ-05 is the producer QC primitive, not an assertion of this profile's outcome. Its truth-domain rule is exact: **T** only when SCI-PTC authoritatively reports `computed_published`, the bound bytes are readable under the declared decoder, the decoded scalar and unit are exactly (1,1), the publication digest matches,

and the broadcast binds every and only exact member of $\mathcal{D}_{g,o}$; **F** only when a complete authoritative check reports a readable non-one value, wrong unit, digest failure, corruption, or incomplete/extra/duplicate broadcast membership; **U** when payload, decoder, digest, complete check, or required membership input is absent; and **C** when authoritative payload or membership records conflict or lie outside the declared generation domain.

4.4 Restriction set

After structural binding and applicability are established, the following restrictions are jointly required. Every restriction and structural invariant is nonexceptionable.

ID	Required permission predicate	Scientific role and false state
UQ-01	Resolved selection is exactly <code>SCI-PTC:uniform_constant@draft-0.1</code> , with no numerical parameter or override.	Prevents aliasing another constant or family. A known mismatch is ineligible; missing/conflicting selection is structurally unavailable.
UQ-02	Given an already established exact g, p, o and CAL-parent binding, the SCI-PTC compatibility record explicitly permits that coefficient generation for that product generation.	Prevents cross-generation borrowing after a valid question exists. Explicit incompatibility is ineligible; a missing or conflicting identity/binding is structural and decision unavailable.
UQ-03	$\mathcal{D}_{g,o}$ exactly equals the bound PTC output support, is finite and nonempty, and contains no duplicate or ambiguous identity.	A known empty population prevents realization; changed or conflicting population requires a new generation and makes the attempted binding unavailable.
UQ-04	The raw scalar is exactly dimensionless $c_\star = 1$ and the operator is exactly $\mathcal{N}_{\text{mean1}}$ on $\mathcal{D}_{g,o}$.	A different constant, unit, operator, or population is another family/version; known substitution is ineligible.
UQ-05	The payload is published/readable under typed availability, decodes exactly to one, and broadcasts to every and only member identity.	A readable non-one payload is producer-invalid and ineligible. Missing or unreadable payload is unresolved/unavailable and is never replaced by unity.
UQ-06	The requested named consumer has explicit permission in this exact family/version record.	Missing MAP permission does not affect JINC and vice versa. Authoritative absence is consumer-specific ineligibility; missing or conflicting permission is decision unavailable.
UQ-07	The coefficient generation and profile evaluation are immutable, one-way linked, and internally consistent; no earlier product or generation was overwritten.	Known lifecycle violation is ineligible; incomplete lineage is structurally unavailable.

PTC post-fit diagnostics, fitted loadings, sample validity, output retention, coordinate validity, MAP admission, JINC admission, and upstream response or covariance availability are not restrictions of this coefficient-fidelity profile. Their exact states and causes remain visible. Response and uncertainty availability have the owner-supplied role **advisory**: they contribute neutral permission and cannot determine coefficient eligibility. No identity response or zero uncertainty is substituted.

The realized `SCI-PTC:uniform_coefficient_handoff@draft-0.1` evaluation is the complete PTC coefficient-use QC record. No second circular producer-QC-passed predicate is required or permitted. UQ-05 supplies the independently evaluated producer payload fact; the profile combines it with UQ-01–UQ-04 and UQ-06–UQ-07 under VAL’s knowledge algebra.

5 Proposed boundary/profile successors

5.1 MAP

The proposed boundary identity is `SCI-PTC_T0_SCI-MAP_BOUNDARY v0.2-draft.1/r0.1`. It changes only the coefficient facet needed for this family: a selected exact `SCI-PTC:uniform_constant@draft-0.1` payload may occupy

the existing coefficient slot only after an approved family record explicitly grants SCI-MAP permission and the exact `SCI-PTC:uniform_coefficient_handoff@draft-0.1` decision is requested, applicable, eligible, and realized. The payload is dimensionless $\gamma_i = \omega_i = 1$ on the matching occurrence. Every frozen MAP identity, signal, retention, coordinate, response, uncertainty, exposure, influence, admission, placement, membership, contribution, failure, and classification rule remains unchanged.

MAP must verify exact family, version, generation, parent, occurrence, support, selection, profile, and permission binding, retrieve the authorized payload, and classify it independently. Only a finite strictly positive value may enter its existing arithmetic. The family does not make an occurrence MAP-admissible and defines no MAP-local factor. Until the family, profile, boundary, and exact bindings are adopted and active, the route is unavailable.

5.2 JINC

The proposed boundary identity is `SCI-PTC_TO_SCI-JINC_BOUNDARY v0.2-draft.1/r0.1`. It changes only the JINC-facing PTC coefficient facet: the same selected `SCI-PTC:uniform_constant@draft-0.1` payload may enter only after an approved family record separately grants SCI-JINC permission and the exact `SCI-PTC:uniform_coefficient_handoff@draft-0.1` decision is requested, applicable, eligible, and realized. JINC independently classifies $\omega_i = 1$ and, after all existing JINC gates, applies Equation 7. This creates no PTC ownership of κ_{ip} , signed normalization, conditioning, support, or bundle semantics and no base-v0.1 JINC response, covariance, or uncertainty product. All other frozen JINC constraints remain unchanged. Until separate JINC permission and every exact binding is active, the route is unavailable.

5.3 No inference or fallback

MAP permission never implies JINC permission and JINC permission never implies MAP permission. Neither consumer may infer this family from its unit, the numerical value one, another consumer's availability, a label, or a missing selection. Absent selection without an authorized mode default, unregistered family, incompatible generation, missing payload/QC, or missing permission makes the route unavailable. There is no unity, alternate-family, neighboring-product, direct-CAL, no-op-PTC, reconstruction, or generation-repair fallback. PTC-disabled operation remains the RTC-terminal export route and creates no PTC, MAP, or JINC product.

6 NOI compatibility limit

This proposal leaves the selected NOI design law unchanged. On an exact MAP-admitted population, Equation 8 follows only because the proposed PTC coefficient is exact unity. G_{pi} , the admitted population, network stratification, MAP plan, and all NOI balance meaning retain their existing owners.

NOI may use the route only if its required parent record binds the exact MAP product/generation, G_{pi} , `SCI-PTC:uniform_constant@draft-0.1`, coefficient generation and payload, exact parent a_{pi} , and a canonical lossless rational decoding exactly to the MAP value. Unity itself has canonical rational form $1/1$; this does not prove that arbitrary G_{pi} or a_{pi} has a required lossless rational source. If any exact parent binding is absent, NOI design resolution remains unavailable. This proposal does not register a PTC NOI-balance family, authorize a JINC-hosted NOI route, change MAP arithmetic, or approve any of NOI's four pending profile successors.

7 Numerical construction

7.1 Occurrence index and broadcast

Let p identify one exact realized PTC transformed-product/application generation for one observation o . Let g identify a new immutable coefficient generation bound to p , its immutable CAL parent, and the requested, effective-policy, observation-resolved, and realized family selection states. The logical coefficient index is the exact detector-time occurrence i . A compact representation may store one scalar only if it binds the lossless relation

$$\mathcal{B}_g(1) = \{(\iota_i, 1) : i \in \mathcal{D}_{g,o}\}. \quad (1)$$

Time, row, column, shape, cardinality, detector label, coordinate equality, or container position cannot establish broadcast membership.

7.2 Support and population

The proposed coefficient support and normalization population are identical:

$$\mathcal{D}_{g,o} = \mathcal{S}_{g,o}^{\text{coef}} = \mathcal{S}_{p,o}^{\text{PTC,out}}. \quad (2)$$

The family covers every and only exact occurrence in the PTC output support of its bound generation for one observation. The population spans all detectors, times, groups, segments, arrays, and networks represented in that support; none defines a separate normalization population. Observations and product generations are never pooled.

Equation 2 does not make sample validity a support predicate. An occurrence may carry coefficient one while its signal, coordinate, retention, or consumer admission is independently unusable. An occurrence outside the support has no coefficient under this family; absence is not zero.

7.3 Fixed constant and normalization

For finite nonempty $\mathcal{D}_{g,o}$, define

$$c_\star \equiv 1, \quad [c_\star] = 1, \quad (3)$$

$$\tilde{\omega}_i = c_\star, \quad i \in \mathcal{D}_{g,o}, \quad (4)$$

$$m_g = \frac{1}{|\mathcal{D}_{g,o}|} \sum_{j \in \mathcal{D}_{g,o}} \tilde{\omega}_j, \quad (5)$$

$$\omega_i = \mathcal{N}_{\text{mean1}}(\tilde{\omega}_i; \mathcal{D}_{g,o}) \equiv \frac{\tilde{\omega}_i}{m_g} = 1. \quad (6)$$

The unit 1 is dimensionless. Cardinality and summation operate on exact logical identities; a duplicate identity is a structural conflict rather than another population member.

The family has no configurable numerical parameter. A generic positive raw constant c would give

$$\frac{c}{|\mathcal{D}|^{-1} \sum_{j \in \mathcal{D}} c} = 1,$$

so its scale would be non-identifiable. The fixed value $c_\star = 1$ is therefore part of the family definition. A request carrying any override, including another positive constant, is not an instance of this family and cannot be silently ignored, clipped, or normalized into acceptance.

The explicit normalization declares a stable comparison convention. Changing the operator or population changes scientific identity even though this constant would remain numerically one. A singleton population is sufficient. An empty population has no mean and creates no generation or payload; it is not repaired by an empty pass, zero, or a scalar with inferred empty broadcast.

7.4 Consumer consequences

At the proposed JINC boundary,

$$w_{ip} = \kappa_{ip} \omega_i = \kappa_{ip}. \quad (7)$$

The equality concerns only the PTC factor and transfers no JINC ownership.

For the already selected NOI design on an exact MAP-admitted contribution population,

$$a_{pi} = G_{pi} \gamma_i = G_{pi}, \quad B_d = \sum_p \sum_{\substack{i \in C_p \\ \text{detector}(i)=d}} a_{pi}. \quad (8)$$

This does not define G_{pi} , change MAP arithmetic, or change the NOI design law.

8 Assumptions, decision semantics, and lifecycle

8.1 Scientific assumptions and limits

This definition assumes only that one exact realized PTC product generation exists for one observation, its exact output support can be identified, and the support is finite and nonempty. It assumes no statistical distribution, stationarity, independence, homoscedasticity, detector equivalence, exposure equivalence, residual model, sensitivity model, or response model. It neither estimates nor ranks data quality.

The coefficient value is an exact policy constant and therefore introduces no sampling uncertainty. That statement is limited to the coefficient itself. It does not assign zero to calibration, signal, response, covariance, cross-observation, support-selection, family-selection, representation, or performance uncertainty. Unknown remains unknown and absent covariance never means independence.

8.2 Knowledge and four independent axes

Every applicable fact and profile predicate uses $\{\mathbf{T}, \mathbf{F}, \mathbf{U}, \mathbf{C}\}$ with the admitted VAL meanings. Silence is $\mathbf{U}$. A structural contradiction blocks establishment of the question. A non-gating $\mathbf{C}$ is preserved and composes as unresolved $\mathbf{U}$. One decisive $\mathbf{F}$ yields ineligible even when another non-gating predicate is unresolved; without a false, any required unresolved predicate yields decision unavailable; all required predicates true yields eligible. Every cause is retained without evidence-erasing short circuit.

The SCI-PTC:uniform_coefficient_handoff@draft-0.1 record carries independent axes:

- R = requested or not requested. A family not selected and handoff not requested has no eligibility evaluation.
- A = applicable, inapplicable, or applicability unknown under the profile definition. Known outside-support occurrences have $E = \emptyset_E$.
- E = eligible, ineligible, decision unavailable, or uninstantiated $\emptyset_E$ where no question is evaluated.
- Z = realized, incomplete, failed, or not produced for the profile evaluation. Coefficient payload availability remains a separate typed fact.

A realized decision-unavailable record is a successful account of missing authority. If the evaluation artifact itself is incomplete, failed, or not produced, it cannot manufacture an eligibility assertion.

8.3 Payload order and failure scope

Processing first establishes identity, parentage, profile authority, applicability, and producer validity; then excludes producer-invalid payloads; then checks an admitted required payload against exact unity. Finiteness or a numerical value of one cannot supply missing authority, validity, support, or permission. MAP and JINC subsequently perform their own positive/zero/ negative/non-finite/unrepresentable classification.

Condition	Required result	Failure scope
Missing/conflicting Registry, source, parent, occurrence, generation, support, or broadcast identity/binding	Applicability unknown or decision unavailable; no fallback payload	Entire affected coefficient generation
Known empty $\mathcal{D}_{g,o}$	No family instance or payload; no eligibility claim	Observation/generation
After structural identity is established, explicit incompatible-generation permission, wrong selected family, wrong operator/unit, or readable non-one payload	Ineligible with all causes; no reinterpretation	Entire scalar-broadcast generation

Condition	Required result	Failure scope
Missing, unreadable, or unrepresentable scalar/broadcast	Typed availability is unavailable or invalid as appropriate; decision unavailable or failed realization	Entire scalar-broadcast generation
Absent explicit consumer permission	Ineligible for that named consumer only	Consumer-specific route
Occurrence authoritatively outside support	Inapplicable, $E = \emptyset_E$, and no coefficient	Occurrence
Consumer numerical rejection of an otherwise faithful payload	Preserve PTC record; consumer excludes or fails under its rule	Occurrence and consumer
Unrelated metadata/advisory diagnostic missing or conflicting	Preserve state and cause; neutral for this profile	No coefficient-QC failure

8.4 Immutable lifecycle and repetition

Creating g never changes transformed product p or an earlier coefficient generation. A change in support, product/application generation, observation, family/version, operator, population, unit, profile, permission, or selection lineage creates a new scientific identity and preserves all earlier parents.

Repeated realization from the same exact inputs must yield the same mathematical payload and membership relation, but each realization retains its own immutable generation identity. Equal values do not authorize substitution or borrowing. Cross-generation use requires an explicit compatibility record binding both generations and preserving their claims. This proposal provides no general compatibility alias.

9 Scientific validity and uncertainty status

Claim or uncertainty	Status class	Meaning in this draft
Frozen ownership and generic coefficient constraints	Approved/recovered	Adopted from the admitted packet and not reopened.
Uniform definition, normalization, support, and profile	Unresolved decision	Complete author proposal awaiting owner approval.
Separate MAP and JINC permissions	Unresolved decisions	Proposed for both; each remains independently open.
Algebraic positivity and mean-one identity	Engineering conformance claim	Follows analytically for exact one on a finite nonempty population; eventual artifact tests must confirm exact realization.
Coefficient sampling uncertainty	Known by definition	None is introduced by the fixed constant; this is not a zero estimate of physical uncertainty.
Family- and support-selection uncertainty	Known but not supplied / unresolved	Not encoded in the coefficient. Unknown is not zero.
Signal, calibration, response, covariance, and cross-observation uncertainty	Producer facts, possibly unavailable	Preserved without reinterpretation; absence does not imply zero or independence.
Implementation conformity	Implementation unassessed	No implementation was inspected and no conformance claim is made.
Representation and response fidelity	Validation not performed	Exact encoding, broadcast, lineage, and whole-chain behavior require later evidence.
Observational performance and comparative suitability	Validation not performed	No sensitivity, variance, bias, or optimality claim is made.
Production readiness	Validation not performed	No route is authorized by this draft.
NOI profile successors	Deferred/unavailable	All four retain their admitted proposed, unapproved, Registry-unbound, and unevaluable status.

10 Decision boundary

The draft has no blocked mathematical definition. Every consequential new choice remains an open author proposal in the owner-decision ledger. Until those choices are adopted and exact successor source/Registry bindings exist, the family, profile, permissions, and boundary successors remain unavailable for evaluation or use. Approval of this family would not close deferred families, mode defaults, or independently governed route prerequisites.

11 Compact owner-decision register

This common-source register is explicitly checked against the companion `OWNER_DECISION_LEDGER.md`; the ledger carries the full owner, evidence, resolution-authority, date, and affected-document fields.

Recovered and deferred states

IDs	State	Disposition
UFC-REC-001–007	decided	Admitted PTC, registry, MAP, JINC, NOI, VAL, and PTC-disabled-route facts are recovered constraints; no claim is blocked.
UFC-DEF-001–005	deferred	Other coefficient families, defaults, NOI profile successors, performance criteria, and a general cross-generation alias remain outside this draft.

Open author proposals

ID	State	Exact blocked claim or output
UFC-OD-001	open	Registered exact-unity numerical family and payload.
UFC-OD-002	open	Exact logical index and scalar-broadcast declaration.
UFC-OD-003	open	Approved support and normalization population.
UFC-OD-004	open	Approved mean-one operator and empty/singleton behavior.
UFC-OD-005	open	Approved no-parameter selection rule.
UFC-OD-006	open	Complete evaluable coefficient-handoff profile identity, use, object, and applicability.
UFC-OD-007	open	Approved UQ-01–UQ-07 truth rules, no-exception rule, advisory roles, and failure scopes.
UFC-OD-008	open	SCI-MAP permission and MAP use of this family.
UFC-OD-009	open	SCI-JINC permission and JINC use of this family.
UFC-OD-010	open	Immutable MAP and JINC successor-boundary bindings.
UFC-OD-011	open	Exact actual-owner field for active Registry/profile binding; the mathematical draft is not blocked.
UFC-OD-012	open	Approved generation/replay and cross-generation rules.
UFC-OD-013	open	Approved coefficient uncertainty meaning.
UFC-OD-014	open	Active coefficient Registry identity and source binding.

Precise open question

ID	State	Exact question and blocked output
UFC-Q-001	open	Is the actual scientific owner Grant Wilson acting for SCI-PTC, as proposed, or another named authority? This blocks active Registry/profile binding only.

12 Normative proposed requirements

Every requirement below is a proposal pending scientific-owner adoption.

ID	Requirement
UFC-REQ-001	The family shall have exact identity <code>SCI-PTC:uniform_constant@draft-0.1</code> , scientific owner SCI-PTC, and no active or registered status until owner approval and exact Registry binding.
UFC-REQ-002	The logical index shall be exact detector-time occurrence i ; any scalar storage shall bind the complete lossless broadcast relation in Equation 1.
UFC-REQ-003	For one observation and bound PTC product generation, coefficient support and normalization population shall both equal the exact PTC output support as in Equation 2.
UFC-REQ-004	On a finite nonempty population, the raw constant shall be exact dimensionless one and normalization shall be the arithmetic mean-one operator in Equations 3–6.
UFC-REQ-005	The family shall accept no numerical parameter. Any constant, unit, operator, or population override shall require another registered family or version and shall not be silently ignored.
UFC-REQ-006	A singleton population shall be valid. An empty, duplicate, ambiguous, or conflicting population shall create no realized coefficient generation and shall not be repaired by zero, unity, deduplication, or an empty pass.
UFC-REQ-007	Occurrences outside coefficient support shall have no coefficient under this family; absence shall remain distinct from numerical zero.
UFC-REQ-008	Every generation shall bind exact transformed product, application generation, CAL parent, observation, selection lineage, support, family, profile, and publication identities without mutating any parent or earlier generation.
UFC-REQ-009	Repeated realizations shall reproduce exact values and broadcast membership but retain distinct immutable generation identities; numerical equality alone shall not permit borrowing.
UFC-REQ-010	The family record shall state SCI-MAP and SCI-JINC permissions separately. Each proposed permission requires independent owner adoption and neither shall imply the other.
UFC-REQ-011	Coefficient fidelity shall be evaluated only under the complete proposed <code>SCI-PTC:uniform_coefficient_handoff@draft-0.1</code> record; the unsupported generic coefficient-QC placeholder shall not be used or simulated.
UFC-REQ-012	The profile shall bind exact use, owner, source, object/domain, restrictions, no-exception rule, response/uncertainty roles, compatibility, supersession, missing/conflict behavior, aggregation status, and lifecycle.
UFC-REQ-013	Structural identity, sample validity, payload availability, producer QC, MAP admission, JINC admission, and each consumer's numerical classification and contribution shall remain distinct.
UFC-REQ-014	Every applicable fact and predicate shall use T/F/U/C semantics; silence shall be unknown, conflict shall be preserved, known exclusion shall dominate unrelated unknowns, and no evidence shall be erased.
UFC-REQ-015	Requested, applicability, eligibility, and realization shall be four independent axes, including uninstantiated eligibility where no question is evaluated.
UFC-REQ-016	All profile restrictions and structural invariants shall be nonexceptionable. No exception from VAL, another use, consumer, or mode shall alter them.
UFC-REQ-017	Missing, invalid, conflicting, unavailable, exact zero, non-finite, and unrepresentable states shall remain distinct and shall follow the stated failure scopes.
UFC-REQ-018	Response and uncertainty availability shall be advisory for this coefficient-fidelity profile; exact upstream states shall be preserved and neither identity response nor zero uncertainty shall be inferred.
UFC-REQ-019	The coefficient shall be nonprecision and shall not be interpreted as inverse variance, sensitivity, significance, fitted loading, scatter, validity, exposure, response, covariance, independence, or optimality.
UFC-REQ-020	The MAP successor shall change only its coefficient binding, retain all MAP-owned arithmetic and gates, and require MAP's independent finite-strictly-positive classification before use.

ID	Requirement
UFC-REQ-021	The JINC successor shall change only its PTC-coefficient binding, retain JINC ownership of κ_{ip} and its arithmetic/gates, and create no new JINC response, covariance, uncertainty, or bundle role.
UFC-REQ-022	NOI compatibility shall require the frozen exact lossless rational-parent record for the exact MAP and coefficient generations. Missing binding shall leave NOI unavailable and shall not change its design law.
UFC-REQ-023	No family or mode default shall be inferred or proposed. Missing selection or authority shall make the route unavailable without fallback.
UFC-REQ-024	Every material change of definition, support, operator, population, unit, profile, permission, boundary, or missing/conflict behavior shall create a versioned successor and shall not rewrite an earlier evaluation.
UFC-REQ-025	Algebraic conformance, implementation conformity, representation/response fidelity, observational performance, and production readiness shall remain separate evidence claims.

13 Predictions and discriminating edge cases

These predictions follow from the proposal; they do not claim that a system has been tested.

ID	Prediction or edge case
UFC-PRED-001	For every finite nonempty exact population and every member, $\tilde{\omega}_i = \omega_i = 1$ exactly; every pairwise coefficient ratio and the arithmetic mean are one.
UFC-PRED-002	A population with exactly one occurrence is valid and produces one exact coefficient.
UFC-PRED-003	An empty population produces no mean, scalar broadcast, or realized family payload.
UFC-PRED-004	Splitting, merging, relabeling, adding, or removing groups, segments, arrays, networks, detectors, or times leaves member values at one when support is otherwise valid, but any support change creates a new generation identity.
UFC-PRED-005	Repeated realizations from the same exact inputs are numerically identical, yet substituting one generation for another without an explicit compatibility record is forbidden.
UFC-PRED-006	A request containing $c \neq 1$, even when $c > 0$ and mean normalization would cancel it, does not select this family and cannot be accepted by silently dropping the parameter.
UFC-PRED-007	A readable value other than exact one fails producer fidelity. Exact zero, a negative finite value, a non-finite value, or an unrepresentable value also fails the consumer's independent positive classification.
UFC-PRED-008	A missing payload cannot be replaced by one even though one is the defining constant; the route remains unavailable.
UFC-PRED-009	A scalar one without an exact occurrence-member broadcast is structurally insufficient and yields decision unavailable.
UFC-PRED-010	An invalid signal occurrence can still have a faithful coefficient one. It contributes only if all independent producer and consumer gates admit the signal and occurrence.
UFC-PRED-011	MAP consumption gives $\gamma_i = 1$ and removes only relative variation attributable to the PTC coefficient; it does not make total MAP contributions equal.
UFC-PRED-012	JINC consumption gives $w_{ip} = \kappa_{ip}$ for the PTC factor, including the sign supplied by JINC; it does not make JINC weights positive or uniform.
UFC-PRED-013	MAP permission alone leaves the JINC route unavailable, and JINC permission alone leaves the MAP route unavailable.
UFC-PRED-014	With no selected family and no authorized mode default, neither consumer may infer this family from a stored one, a label, or another route.
UFC-PRED-015	On an otherwise valid exact MAP host route, $a_{pi} = G_{pi}$; missing lossless rational identity for that exact a_{pi} still leaves NOI design resolution unavailable.
UFC-PRED-016	PTC-disabled or missing positive-rank PTC realization produces no instance of this coefficient and creates no MAP or JINC route.

14 Evidence required after owner adoption

Eventual conformance evidence must bind the exact approved source and Registry records. Algebraic tests must cover nonempty populations across detector, time, group, segment, array, and network labels; singleton support; exact mean-one results; and exact scalar broadcast membership. Negative tests must cover empty, duplicate, ambiguous, or changed support; parameter overrides; wrong unit/operator/family; missing or non-one payload; unreadable or unrepresentable value; wrong product or generation; absent consumer permission; and cross-generation borrowing. Replay evidence must show exact numerical reproducibility while retaining distinct immutable generations.

Profile evidence must exercise T/F/U/C composition, all four decision axes, nonexceptionability, structural versus non-gating conflict, payload ordering, cause preservation, and every failure scope. Boundary evidence must show MAP and JINC performing independent numerical classification and admission, permissions not crossing, and no fallback route. NOI evidence, if that independently governed route becomes evaluable, must bind the exact MAP value and canonical lossless rational parent without changing MAP arithmetic.

Scientific validation is a separate future activity. It requires owner-approved observational or simulation populations, comparison questions, metrics, uncertainty treatment, and acceptance criteria. None is invented here. Such evidence could establish comparative performance or suitability; algebraic tests cannot. Production readiness adds operational completeness and failure-handling evidence only after scientific and implementation conformance are established.